JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to THIRUKUMARAN PRAVEEN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

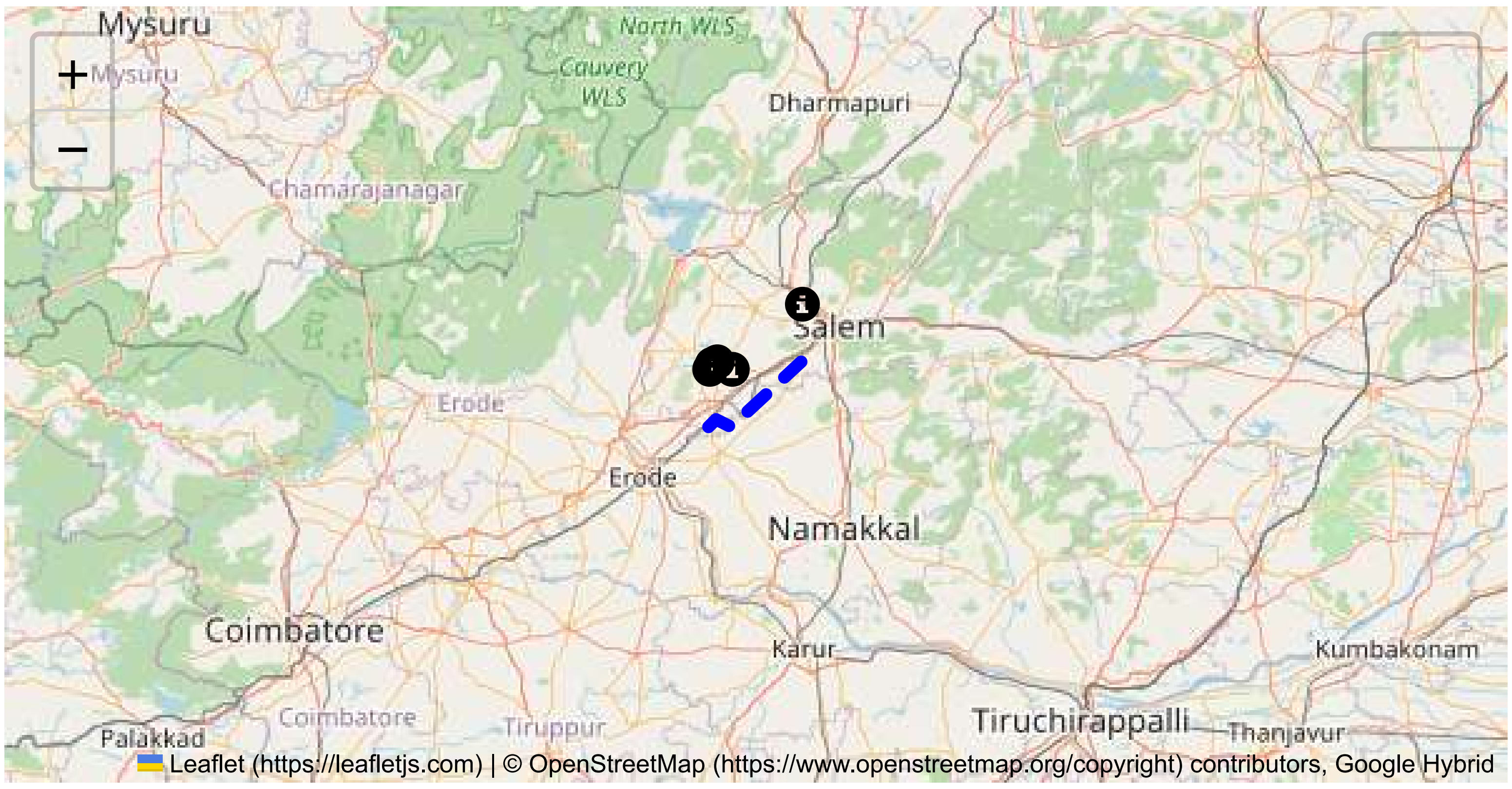
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

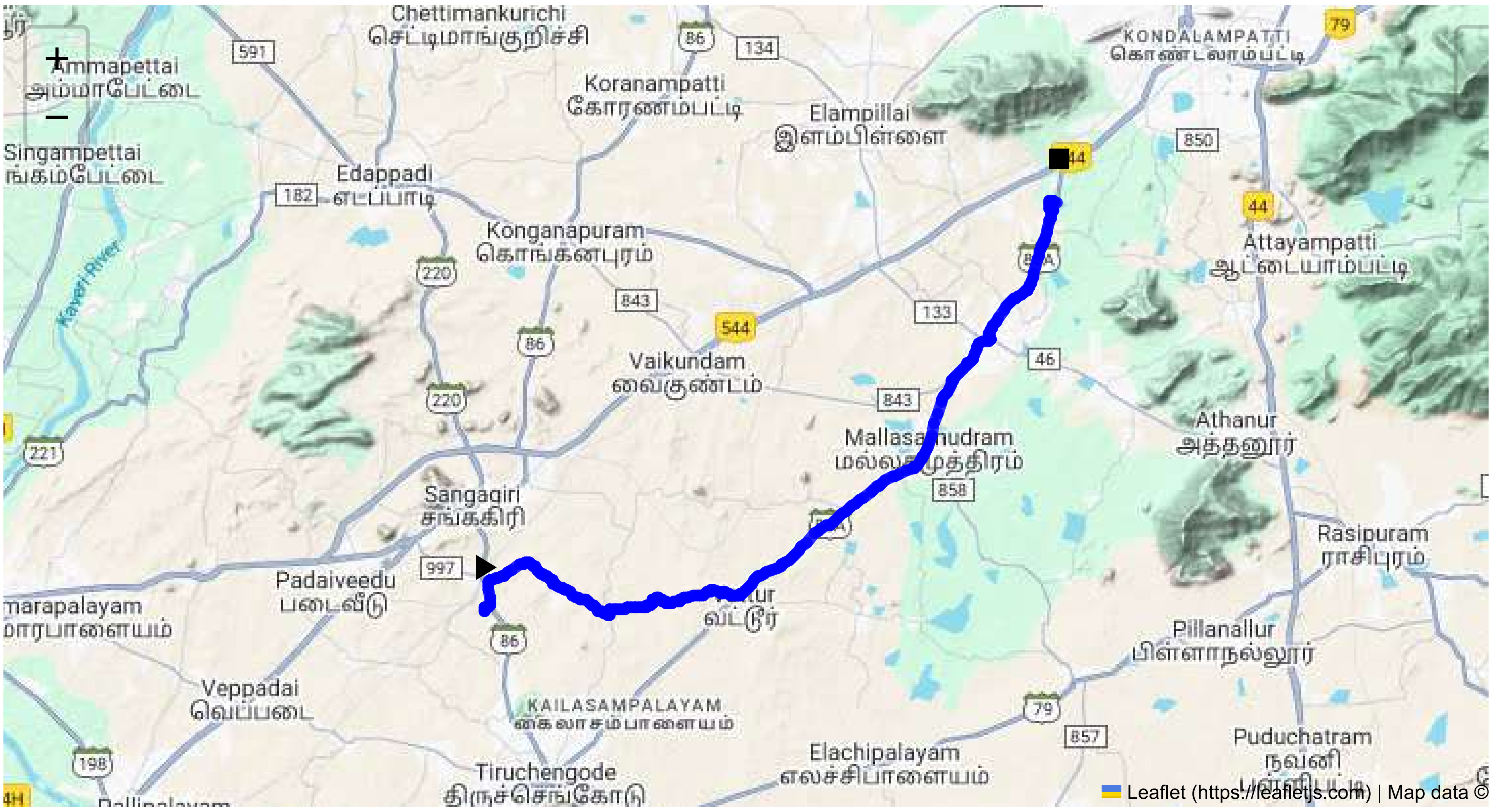
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 33.49 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.57907974, 78.07425984)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 33.49 kilometers and passes through several local roads starting from Sangagiri in Tamil Nadu to Veerapandi, Akkarapalayam. The key waypoints include P-62, Sankari R.S, and Kozhikkalnatham. The route progresses through mainly rural and semi-urban areas, with segments that traverse agricultural and residential regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate, with hot, humid summers and significant monsoon rainfall from June to September. During the monsoon, roads can become slippery, and localized flooding may occur, particularly in low-lying areas. Heavy fog can appear occasionally in the winter months, reducing visibility.

3. Analysis of Traffic Patterns

Traffic congestion is typically moderate, with peak hours in rural areas generally occurring in the early morning (7-9 AM) and late afternoon (5-7 PM). Town centers and market areas along the route are prone to congestion, especially during local festivals or market days.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route. Some segments, particularly those near Sangagiri, are well-maintained with clear markings. However, as you proceed into rural areas, road conditions can deteriorate, with potential for narrow lanes, potholes, and lack of adequate signage. Infrastructure such as shoulder lanes can be minimal, increasing risk for heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies

Alternative routes may include state highways or main roads that bypass heavily congested or poorly maintained sections. In case of emergencies, considering nearby highways like NH 544 or SH 96 might provide quicker access to primary towns or cities for assistance.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Local regulations require that hazardous materials be transported with appropriate signage and containment to prevent spills. Nighttime movement might be discouraged in certain stretches due to reduced visibility and lack of support infrastructure. It is mandatory to follow all safety protocols, including having spill kits and fire extinguishers on board.

7. Overview of Historical Incidents

There have been isolated incidents of accidents involving heavy vehicles in the region, generally attributed to poor road conditions or mechanical failures. Hazardous material spills are not common but could have severe impacts due to the proximity to agricultural lands.

8. Environmental Considerations and Sensitive Areas

Sensitive areas include those near water bodies and agricultural fields. Proper care during transit, especially spill prevention, is crucial to protect the local ecosystem. Avoiding wetland areas nearby during monsoon is advised to prevent contamination.

9. Analysis of Communication Coverage

Communication coverage along the route is generally adequate, but there might be occasional dead zones, especially in the rural stretches. It is advisable to have emergency contact numbers pre-loaded and inform local stations of your route and timetable.

10. Estimated Emergency Response Times

Emergency response times may vary from 30 to 60 minutes, depending on the proximity to major towns such as Salem. Rural areas might experience slower response due to limited infrastructure and access.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Veerapandi, Akkarapalayam poses moderate risks for trucks carrying hazardous material, primarily due to varying road conditions and potential weather hazards. Effective risk management entails careful planning around peak traffic hours, heightened awareness of road conditions, and adherence to local transportation regulations. Given the environmental considerations and potential dead zones in communication, proactive measures should include equipping vehicles with emergency and safety tools and maintaining an open line of communication with central dispatch and emergency services. Emergency planning should include route alternatives and real-time weather and traffic updates to navigate safely during adverse conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
5	Turn	Medium	11.44194, 77.90923	30 KM/Hr
6	Blind Spot	Blind Spot	11.43673, 77.91703	10 KM/Hr
7	Turn	High	11.43926, 77.91705	15 KM/Hr
8	Turn	Medium	11.43931, 77.92291	30 KM/Hr
9	Turn	Medium	11.43982, 77.93018	30 KM/Hr
10	Turn	Medium	11.44080, 77.93220	30 KM/Hr
11	Turn	Medium	11.44135, 77.93220	30 KM/Hr
12	Turn	Medium	11.44175, 77.93249	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.44234, 77.93242	30 KM/Hr
14	Turn	High	11.44248, 77.93247	15 KM/Hr
15	Turn	Medium	11.44547, 77.95309	30 KM/Hr
16	Turn	Medium	11.44370, 77.96020	30 KM/Hr
17	Turn	High	11.44387, 77.96283	15 KM/Hr
18	Turn	Medium	11.53168, 78.05072	30 KM/Hr
19	Turn	Medium	11.54862, 78.06397	30 KM/Hr
20	Blind Spot	Blind Spot	11.57645, 78.07293	10 KM/Hr
21	Turn	Medium	11.57706, 78.07138	30 KM/Hr
22	Turn	Medium	11.57712, 78.07135	30 KM/Hr
23	Turn	High	11.57861, 78.07164	15 KM/Hr
24	Turn	Medium	11.57866, 78.07178	30 KM/Hr
25	Turn	Medium	11.57877, 78.07188	30 KM/Hr
26	Blind Spot	Blind Spot	11.57975, 78.07200	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	police	Mallasamudram Police Station	11.4998427, 78.0303028	30 km/h	Medium
6	police	Attayampati C 1 Police Station	11.5313534, 78.050867	30 km/h	Medium
7	hospital	Mythili Hospital	11.533594, 78.050881	30 km/h	Medium
8	clinic	A. Thangavel Physiotherapy Clinic	11.5314071, 78.052862	30 km/h	Medium
9	hospital	Dhanalakshmi Hospital	11.5318866, 78.0514614	30 km/h	Medium
10	hospital	Malligai Nursing Home	11.5308602, 78.0529837	30 km/h	Medium
11	hospital	Nalam Hospital	11.5315309, 78.0533249	30 km/h	Medium
12	hospital	Sree Vasantham Hospital	11.540146, 78.0545787	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium
4	school	Mahendra Matriculation School	11.5120354, 78.0354797	30 km/h	Medium
5	college	Mahendra Arts and science college	11.5112305, 78.0344287	30 km/h	Medium
13	school	Vivekananda Balamandir Matric School	11.5395679, 78.0559776	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44861, 77.87429

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44194, 77.90923

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.43673, 77.91703

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43926, 77.91705

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43931, 77.92291

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43982, 77.93018

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44080, 77.93220

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44135, 77.93220

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44175, 77.93249

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44234, 77.93242

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44248, 77.93247

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44547, 77.95309

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44370, 77.96020

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44387, 77.96283

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53168, 78.05072

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54862, 78.06397

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.57645, 78.07293